DustLedger

Measure each construction site's own dust — so Delhi can act on the dirty sites instead of stopping all of them

Team: ‹TEAM NAME› · **Team ID:** ‹ID› · **Theme 2:** Rising AQI Levels in Modern Urban Cities

‹Member 1 — role› · ‹Member 2 — role› · ‹Member 3 — role› · ‹Member 4 — role› · ‹email / phone›

This winter, Delhi restricts every construction site

When	What
Nov 1 – Jan 31	dust-generating demolition and outdoor civil works **prohibited**
Dec 10 – Jan 20	**total ban** on construction and demolition
GRAP emergencies	stage-wise stops on top; GRAP-3 halted construction for 16 days in 2025 (₹10,000 compensation announced)
Every site	100% CCTV coverage mandated

The same rule for every site — whatever its dust control.

Sources: Delhi Winter Action Plan 2026-27 as reported by The News Mill (Sept 2026, secondary); Hindustan Times (Dec 2025). Full table: research/fact-check.md (F1, F3).

A monitor can say the air is bad — not which site made it bad

- **City-scale sensing:** 47 reference monitors in Delhi (going to 60); planning norm is one per 25-50 km²
- **Site compliance = checklists:** anti-smog guns (₹7,500 fine per missing gun per 5,000 m²), CCTV, self-audit uploads
- **Fines by area, not by dust:** CAQM's 2024 schedule sets ₹1 lakh to ₹2 lakh per construction-site violation depending on area, doubled under GRAP
- **Trust is broken:** in Oct 2025 water tankers were filmed spraying at the Jahangirpuri reference monitor; CPCB says its data can't be manipulated

Penalties track checklists, not the dust a site actually releases.

Sources: Hindustan Times (2026; Feb 2024, reporting the CAQM order of 6 Feb 2024); Newslaundry (30 Oct 2025); New Indian Express (3 Dec 2025). Fact rows F5, F6, F25.

A small source, handled with the bluntest tool

2-5%

of Delhi's ambient PM10 came from construction material in IIT Kanpur's receptor study for DPCC (data 2013–14).
Construction & demolition: 3.6% of PM10 *emissions*; concrete batching another 10%.

1,378,199

construction workers on Delhi's welfare-board rolls — only 89,518 active members.
In one survey, 5 of 75 site workers were registered; none had received the ₹8,000 aid.

Blanket stops cost most to those least able to pay — for a few per cent of the city's PM.

Sources: IIT Kanpur (2016) for DPCC, §4.6 and Table 3.1 (primary; 3.6% is our arithmetic: 5.1668 of 143 t/d); Down To Earth (Dec 2024). Fact rows F7–F10.

Why today's tools fall short

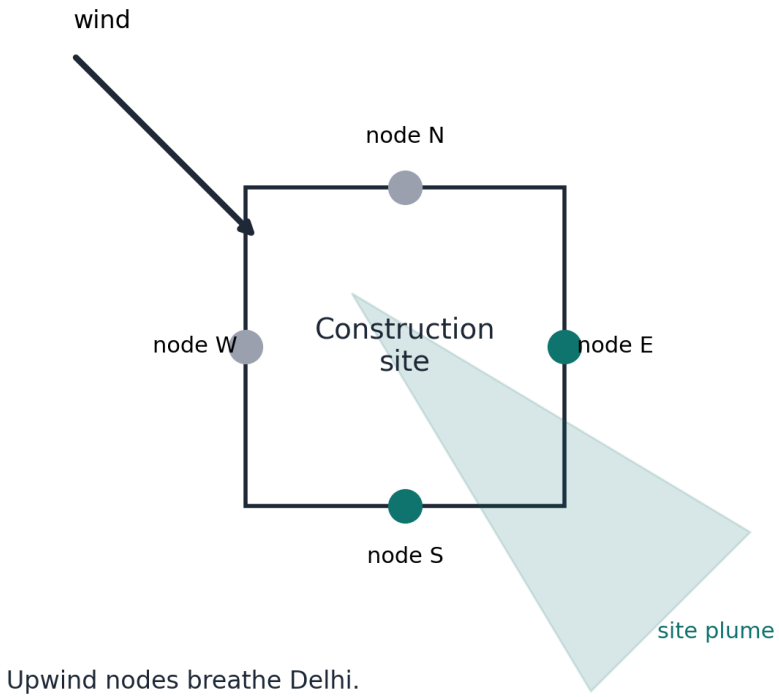
What exists	The gap
Reference monitors (CAAQMS)	city scale — can't single out a site
Compliance monitors streaming to the DPCC cloud (e.g. Airveda, Prana Air)	absolute PM; Delhi's winter PM10 averages 314 ± 77 µg/m³, so a site is a small difference on top
Anti-smog guns, sprinklers	run on timers — no feedback on whether dust fell
City AQI forecasts (IITM, Google)	tomorrow's city AQI, not today's dusty site
Fence-line dust systems (Aeroqual, Envirosuite)	mines and refineries; we found no low-cost Indian construction workflow in the product pages we reviewed

Sources: research/landscape.md §3–4 (vendor pages; TERI–ARAI 2018). Fact row F11.

DustLedger: a fence-line ring that subtracts Delhi

1. **Sense** — 4 PM nodes + 1 wind sensor on the boundary; solar + LoRa; LTE gateway PLANNED
2. **Attribute** — downwind minus upwind = the site's own dust; says **"inconclusive"** in calm or confused wind SIMULATED
3. **Act** — mist only when the site is the cause, flow-metered PLANNED
4. **Prove** — signed, hash-chained records → a ranking that tells inspectors where to look PLANNED

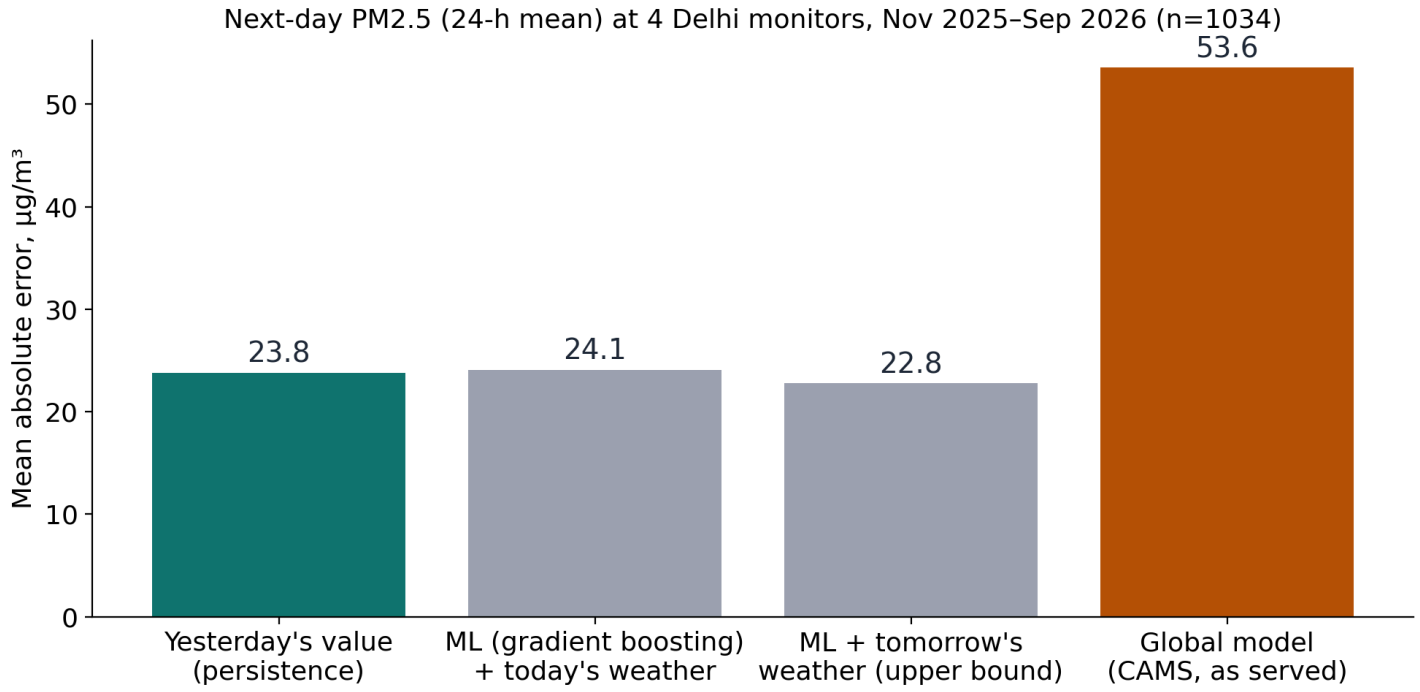
Data pipeline on real Delhi monitors + ERA5 BUILT



Upwind nodes breathe Delhi.
Downwind nodes breathe Delhi + the site.
The difference is the site's own dust.

Status labels: BUILT · SIMULATED · PLANNED · every run number is rendered from committed results (eval/render_docs.py)

Real Delhi data: next-day city forecasting is not the gap BUILT

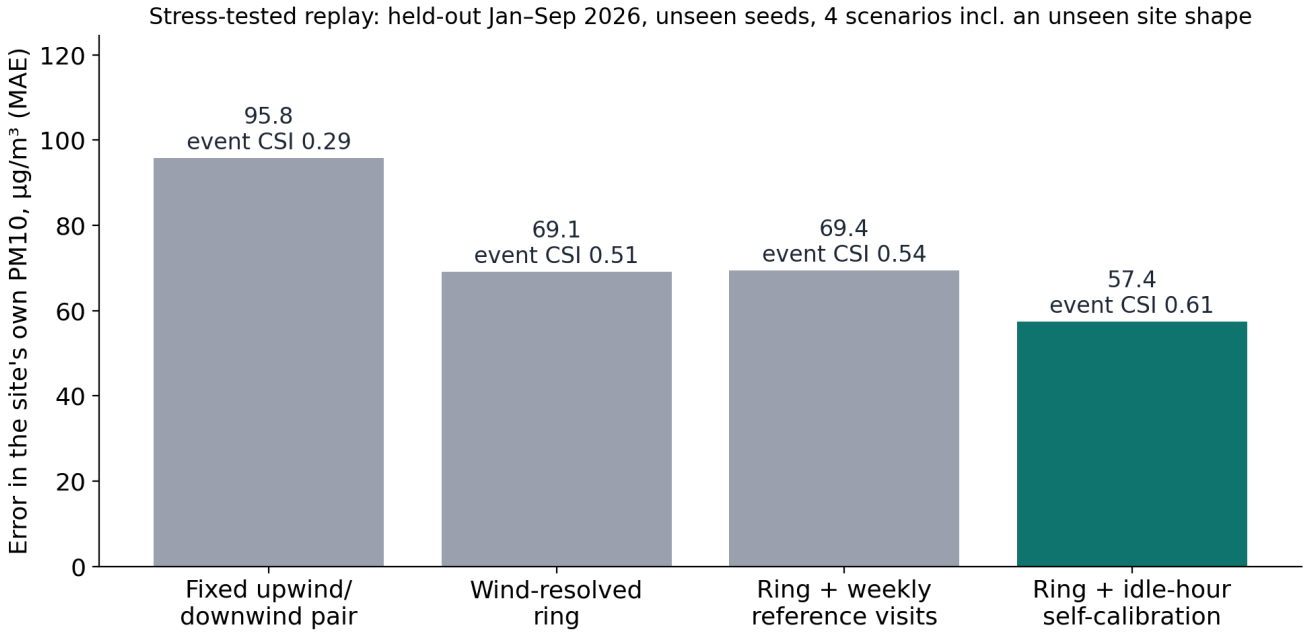


Real CPCB/DPCC data (OpenAQ archive) + ERA5. Walk-forward, time-ordered. eval/forecast_walkforward.py

Yesterday's value predicts tomorrow's PM2.5 within **23.8 µg/m³** on average (r 0.91, n = 1,034 station-days); our ML adds at most 4%. So forecasts only schedule dusty work — the innovation is attribution.

Attribution, stress-tested

SIMULATED site · REAL background + wind



REAL background PM10 (CPCB monitor) + REAL ERA5 wind; SIMULATED site, plumes, sensors and faults. autoresearch/eval_v2.py (v2.1)

The wind-resolved ring cuts the error by 28% vs a fixed pair; re-calibrating nodes against each other in quiet hours cuts another 17% vs weekly reference visits (**simulation only**). In Nov–Jan, ERA5 wind is below 1 m/s in 16.2% of hours — we say "inconclusive" there.

Cost and power — from vendor listings and datasheets

Kit parts: 4 nodes + wind + gateway + misting	**₹60,711** (listed retail parts incl. GST; excludes masts, cabling, data plan, installation, upkeep)
One node	₹10,032
Wind sensor	₹14,810 — 24% of the kit
Power per node	10.2 Wh/day (datasheet currents + stated assumptions)
10 W panel in December (worst month)	2.1× the daily load
Battery with no sun	12.1 days

Yardstick: one construction-site violation costs ₹1 lakh to ₹2 lakh under CAQM's 2024 schedule (doubled under GRAP). The parts are 30–61% of one fine — not a lifetime cost.

Prices seen by the lead on robu.in / amazon.in, 4 Oct 2026 (deliverables/bom/bom.csv). Datasheets: Sensirion SPS30, Espressif ESP32-S3; irradiance: NASA POWER (fact rows F20–F22).

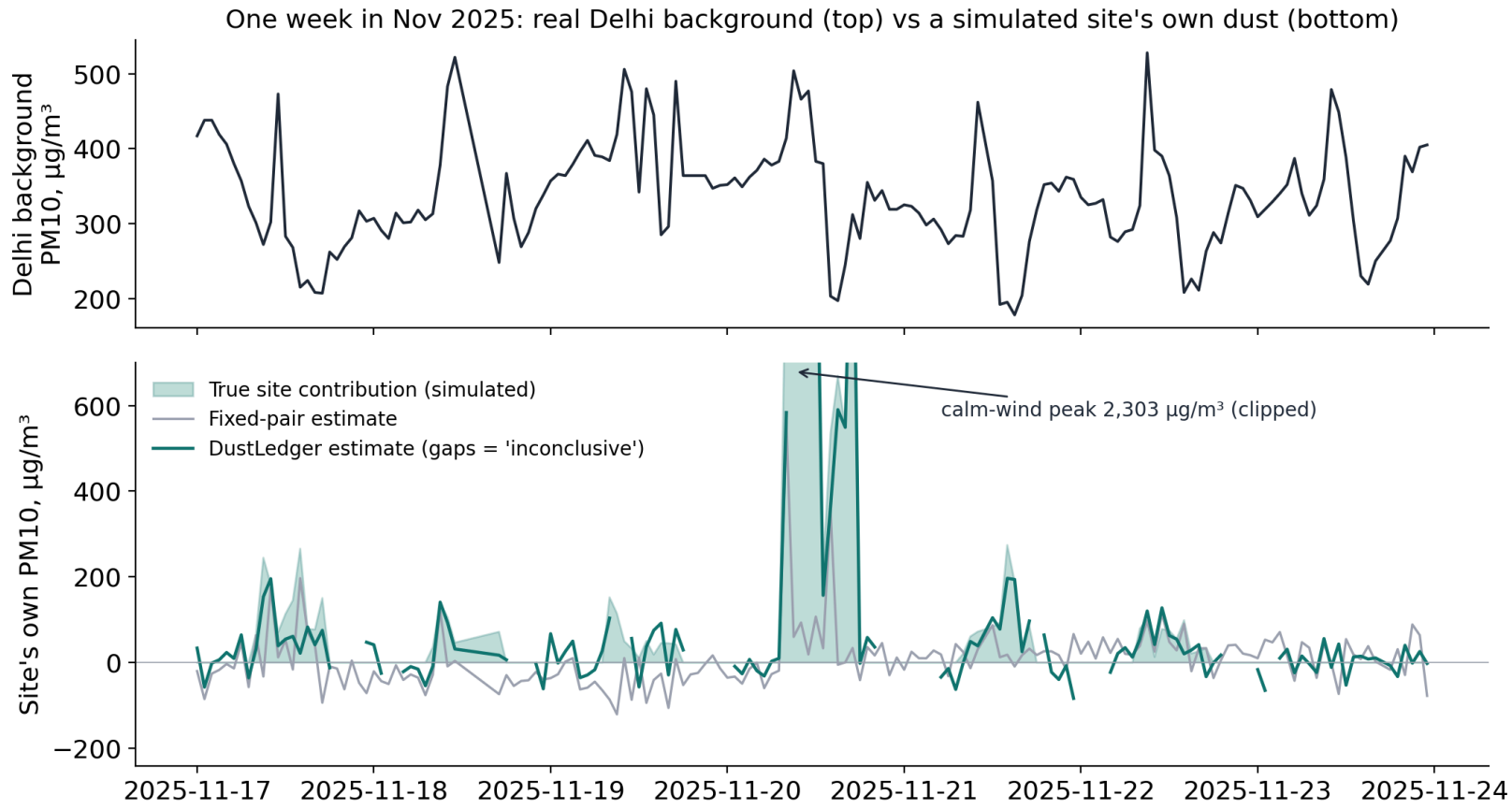
What we will prove by 1 December — and our ask

When	What	Evidence it produces
Oct	order parts; 4 nodes side by side; reference co-location if a slot is available	inter-node agreement; calibration vs a reference
Before 1 Nov	real-site trial window (Delhi's restrictions start 1 Nov) — or a site permitted to operate	first real fence-line increments
Nov	controlled dust-source trial: ring + wind + misting on/off	site increment, abstention rate, litres used
1 Dec	report + working prototype	real sensor data, labelled

Proposal, not a promise: an "earned-permission" pilot for verified-clean sites, for CAQM to consider.
Ask: a pilot site · a reference co-location slot · a DPCC/CAQM contact

Measure the site, not the city.

B1 · One week of the replay SIMULATED site



Background = CPCB/DPCC monitor 6929 (real); wind = ERA5 (real); site + sensors SIMULATED (eval_v2.1, 'medium' scenario, seed 11).

B2 · What is built, simulated and planned

BUILT (committed, reproducible)	SIMULATED (labelled)	PLANNED (Round 2)
CPCB/DPCC hourly PM at 4 Delhi monitors (OpenAQ archive) + ERA5	construction site, emissions, plumes	4-node ring, wind mast, gateway
next-day forecast benchmark, walk-forward split	sensor gains, fouling, drift, faults	misting line with flow meter
attribution replay + frozen evaluators (v1, v2.1) + ledger	neighbour site, road, dirty nights	signed records + hash chain
typed component contracts, verified BOM, power budget	misting effect: **not modelled**	supervisor app, regulator ranking

B3 · Calm wind: when we say "inconclusive"

Delhi, Nov–Jan (ERA5, 2023–2026)	Share of hours
wind < 0.5 m/s	4.0%
wind < 1.0 m/s	16.2%
wind < 1.5 m/s	39.0%
median wind	1.7 m/s

ERA5 is a ~25 km grid that smooths out calms, so a site vane will see **more** calm than this. These shares are lower bounds. "Inconclusive" never counts as clean.

B4 · How the stress test works (evaluator v2.1)

Scenario (held-out Jan–Sep 2026, unseen seeds)	Fixed pair	Ring	Ring + weekly reference visits	Ring + self-calibration
easy	80.1	43.6	41.3	36.5
medium	95.4	57.8	53.1	45.2
hard	101.9	81.8	86.0	66.7
hard, unseen site shape	105.9	93.3	97.2	81.3

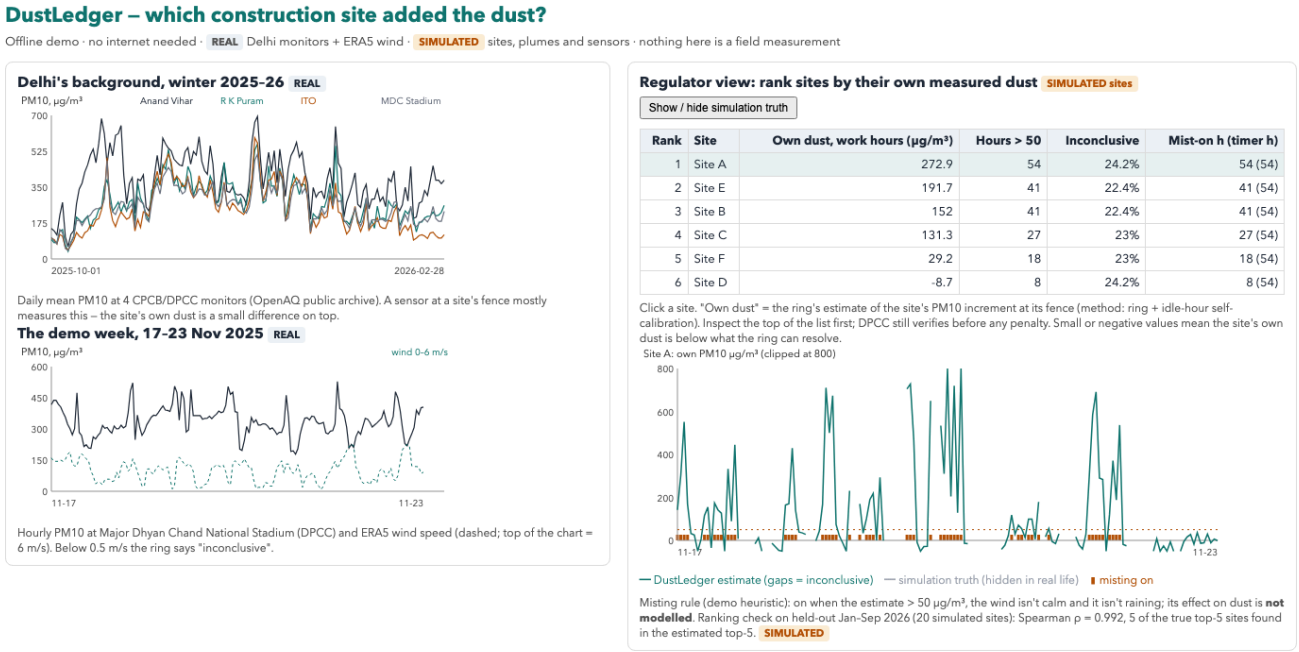
MAE of the site's own PM10, µg/m³. Stress factors from an independent review: separate dust vs ambient sensor gain, fouling steps, offsets, saturation, work on some nights, spatial gradients, a neighbour site, a road, vane bias and decorrelation. Holdout run once by the lead; the earlier, easier v1 simulator overstated the gain.

B5 · Parts list summary (prices seen 4 Oct 2026)

Subsystem	₹ (incl. GST)
4 nodes (SPS30, SHT30, ESP32-S3, SX1262, 10 W panel, LiFePO4 + MPPT charger, IP65 box)	40,128
Wind speed + direction sensor	14,810
Gateway (A7670C LTE, ESP32-S3, SX1262, box)	3,516
Misting kit + flow sensor + relay	2,257
Kit parts total	**60,711**

Not included: buck converters, cabling, masts, radiation shields, SIM/data, installation, reference co-location, upkeep. Every line with its URL: deliverables/bom/bom.csv. The relay listing was out of stock; the SX1262 module's IN865 band is unconfirmed.

B7 · Offline demo (screenshot) SIMULATED sites · REAL background + wind



Six simulated sites ranked by their own dust over a real Delhi week. Held-out check, 20 simulated sites: Spearman ρ 0.99, 5 of the true top-5 found — an easy test (emissions differ up to 30-fold).

B6 · Sources (selected)

- Delhi Winter Action Plan 2026-27 — The News Mill, Sept 2026 (secondary) · GRAP — Hindustan Times, Nov–Dec 2025
- CAQM penalty order (6 Feb 2024) — Hindustan Times, Feb 2024 · CAAQMS network — Hindustan Times, 2026
- Construction workers and aid — Down To Earth, Dec 2024 · Supreme Court order, 25 Nov 2024
- Source apportionment — IIT Kanpur for DPCC, *Comprehensive Study on Air Pollution and GHGs in Delhi*, Jan 2016 (primary)
- Winter PM10 — TERI–ARAI (2018) · Low-cost sensors for construction dust — CSTEP guidance (2025)
- Health — Lancet Planetary Health 10-city study via Hindustan Times (4 Jul 2024)
- Datasheets — Sensirion SPS30 v0.9; Espressif ESP32-S3 v2.1 · Irradiance — NASA POWER climatology
- Data — CPCB/DPCC monitors via the OpenAQ archive; ERA5 and CAMS via Open-Meteo
- Every number with URL, quote and grade: research/fact-check.md · run numbers: eval/render_docs.py